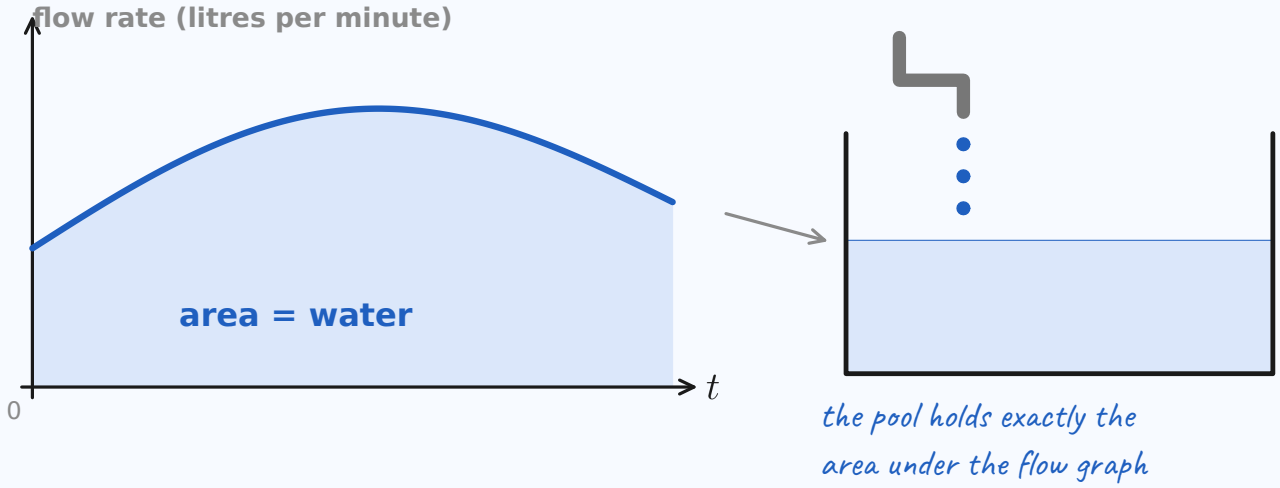
Chapter 36

# Integration and areas

*add up the tiny pieces*



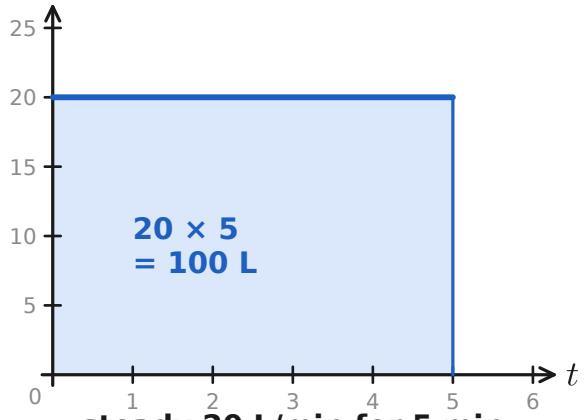

If water pours in at 20 litres a minute for 5 minutes, the pool gains $20 \times 5 = 100$ litres. On a graph of flow rate against time that's a rectangle's **area**. When the rate keeps changing, the water is still the **area under the curve**, and finding it is called **integration**.

It's the second half of calculus, and it turns out to be differentiation run backwards.

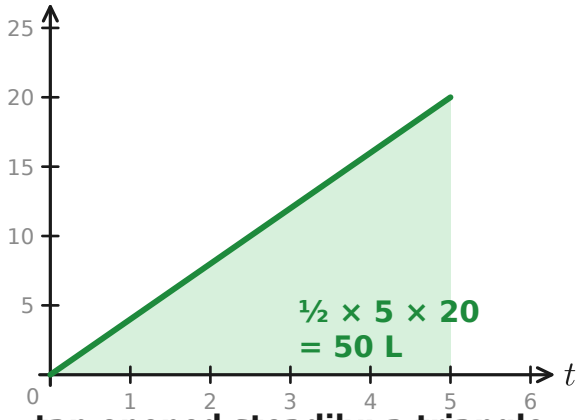
Goes with **Lecture 16** (area under a curve, the trapezium rule and its accuracy, the integral sign, the general formula, +c, 1/x, standard integrals, adding and multiples, definite integrals, negative areas).

# Area under a graph = a total

Rate × time = amount. On a rate–time graph, rate is the height and time is the width, so the amount is an **area**:




**steady 20 L/min for 5 min**



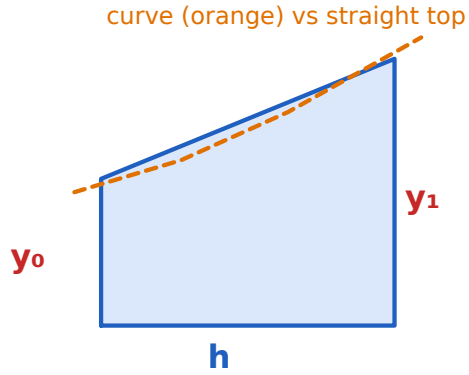

**tap opened steadily: a triangle**

Rectangles and triangles are easy. A **curved** top isn't, which is what the rest of this chapter is for. Lecture 16's examples of "area under a curve" in real life:

- a **probability**: the area under the Normal curve (Chapter 26)
- **displacement**: the area under a velocity–time graph
- **drug absorption** (pharmacokinetics): the area under a concentration curve

# Strips with sloping tops: the trapezium rule

Chop the area into vertical strips of width $h$ and give each strip a straight sloping top. Each strip is a **trapezium** (four sides, two of them parallel). Its area is the **average height × width**:



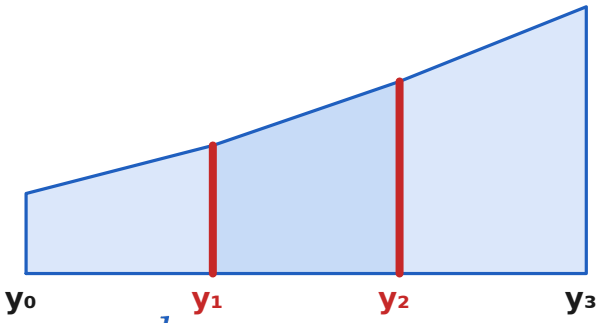

$$\text{area} \approx \frac{y_0 + y_1}{2} \times h = \frac{h}{2}(y_0 + y_1)$$

*the straight top misses the curve a little, so this over- or under-estimates (Lecture 16's "imperfections")*

Lecture 16, one strip: area under $y = x^2$ from $x = 1$ to $x = 4$, so $h = 3$:

$$\text{area} \approx \tfrac{3}{2}\left(1^2 + 4^2\right) = \tfrac{3}{2} \times 17 = 25.5$$

More strips. With several strips side by side, every **inside wall** belongs to two strips (the right side of one, the left side of the next), so it gets counted **twice**:




*red walls $y_1$ and $y_2$ are shared by two strips, so they appear twice. The two end walls $y_0$ and $y_3$ only once.*

$$\text{area} \approx \frac{h}{2}\left(y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n\right)$$

$$h = \frac{b - a}{n}$$ ← *n strips between x = a and x = b*

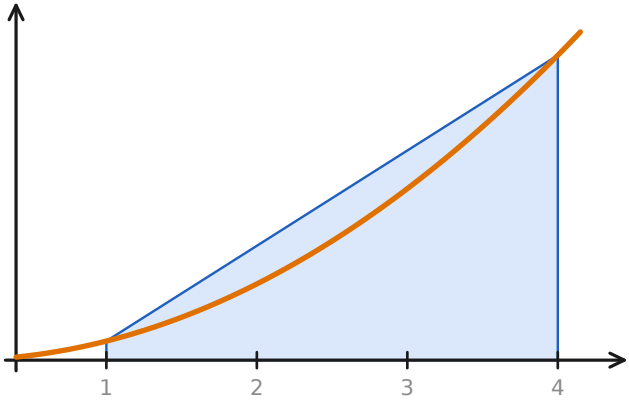
**Slide slip:** Lecture 16's general formula leaves out the 2 in front of the middle heights. Its own two-strip example (next page) does use the 2.

## 1, 2, 4, 8 strips: watch the error shrink

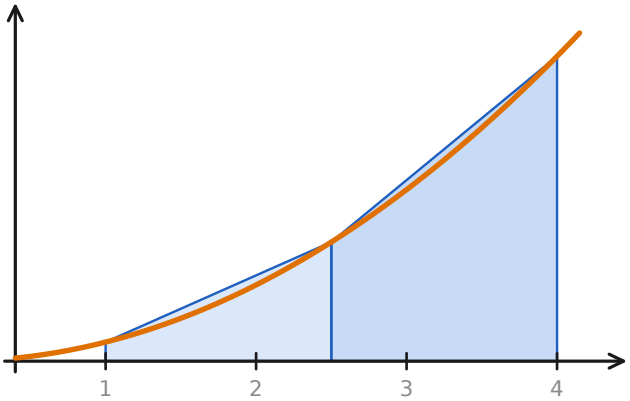
Lecture 16's two-strip version: $h = 1.5$, middle height $2.5^2 = 6.25$:

$$\frac{1.5}{2}\left(1 + 2 \times 6.25 + 16\right) = 0.75 \times 29.5 = 22.125$$

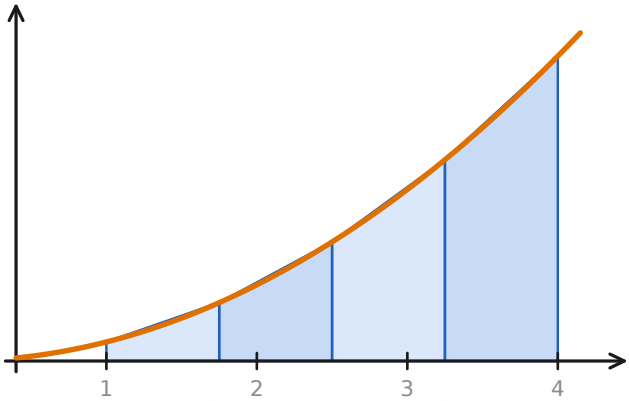
The true area is 21 (you'll prove it later in this chapter). Keep doubling:



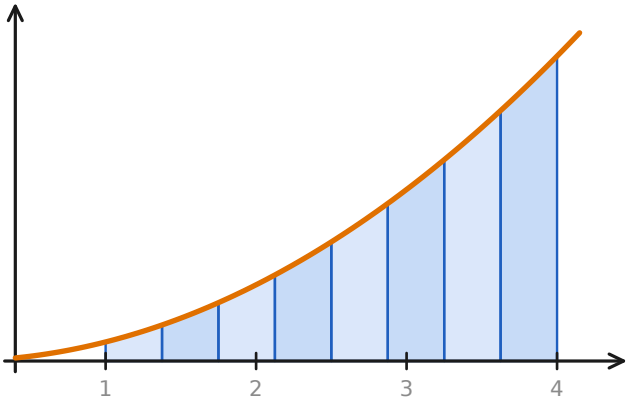

**n = 1**: 25.5 **error 4.5**




**n = 2**: 22.125 **error 1.125**




**n = 4**: 21.2812 **error 0.28125**

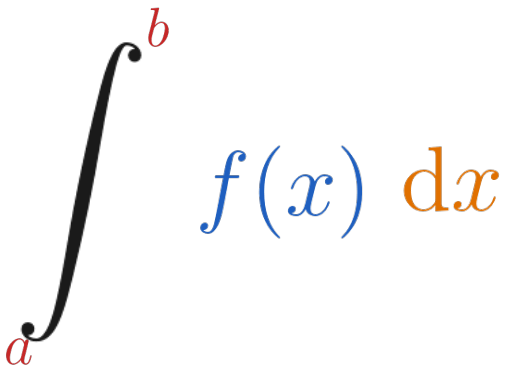



**n = 8**: 21.0703 **error 0.0703125**

Look at the errors: 4.5, 1.125, 0.28, 0.07. For this curve, **halving the strip width cuts the error to exactly a quarter.** For other smooth curves it is only **roughly** a quarter, and only once the strips are fairly narrow; for some curves the pattern is different. What always holds: as the number of strips heads to infinity, the error heads to 0 (Lecture 16: as $n \to \infty$, $\varepsilon \to 0$).

# Infinitely many strips: the integral sign

With a finite number of strips you'd write a sum with $\Sigma$ (Chapter 17). With infinitely many, infinitely thin strips, the S gets stretched into a new symbol, $\int$ (a tall S for "sum"):




$\int$ *"the integral of": add up infinitely many thin strips*

*a and b: the limits (where the area starts and stops)*

*f(x): the integrand, the height of each strip*

*dx: the strip width, shrunk to nothing. It also says which letter you're integrating with.*

## Integration is the undo button for differentiation

**RULE: the reverse power rule**

$\int ax^n \, dx = \frac{a}{n+1}x^{n+1} + c$: add 1 to the power, then divide by the **new** power.

It works for every $n$ **except** $n = -1$, which would divide by 0. That case is $\int \frac{1}{x} \, dx = \ln|x| + c$ (later in this chapter).

A constant: $\int k \, dx = kx + c$ (it is $kx^0$, so the power goes up to 1). For example $\int 5 \, dx = 5x + c$.

Every indefinite integral ends with $+c$. Check by differentiating your answer: you must get back what you started with.

Chapter 14's **undo button**: every operation has an inverse that takes you back. Integration is the undo button for differentiation (Lecture 16): it asks **"what did I differentiate to get this?"**

$$y = 5x^3 + 3 \xrightarrow{\text{differentiate}} 15x^2 \xrightarrow{\text{integrate}} 5x^3 + ?$$

The "+3" vanished when we differentiated, so undoing can't know what it was. More on that in a moment. First the rule. Differentiating made the power **dive down and drop by one**. Undo it in reverse order:

$$x^3 \longrightarrow \frac{x^4}{4}$$

1. the power goes UP by one: 3 → 4

2. divide by the NEW power

$$\int ax^n \, dx = \frac{a}{n+1}x^{n+1} + c \qquad (n \neq -1)$$

Lecture 16: $\int 10x^4\,\mathrm{d}x$.

$$\int 10x^4\,\mathrm{d}x = \frac{10}{4+1}x^{4+1} + c = 2x^5 + c$$

check: $\dfrac{\mathrm{d}}{\mathrm{d}x}(2x^5 + c) = 10x^4$ ✓ ← always check by differentiating back

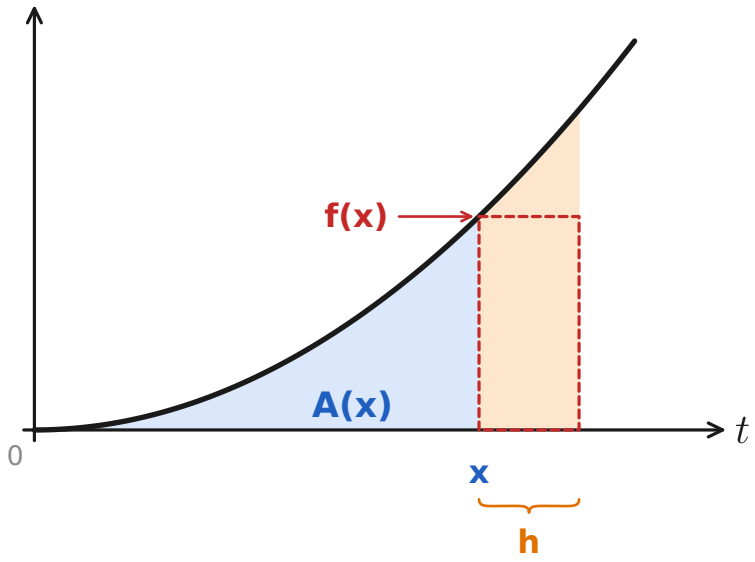
## Why area undoes a gradient: slide the right edge

Why should **area** (adding strips) be the undo of **gradient** (Chapter 34)? Fix the left edge at 0 and let the right edge sit at $x$. Call the shaded area $A(x)$: a number that changes as you slide the edge.

Take $f(t) = t^2$:

✎ **YOUR GO (pencil first)**

*Slide the right edge a tiny distance h further. What shape is the extra sliver of area, roughly? How tall is it, and so what's its area?*




*The new orange sliver is almost a rectangle (red dashes): height f(x), width h.*

*So extra area ≈ f(x) × h.*

*The thinner the sliver, the smaller the curved error on top.*

Turn that into a gradient. The **rise** of $A$ is the sliver, the **run** is $h$:

$$A(x+h) - A(x) \approx f(x) \times h$$ ← extra area = the sliver

$$\frac{A(x+h) - A(x)}{h} \approx f(x)$$ ← ÷ h: rise over run

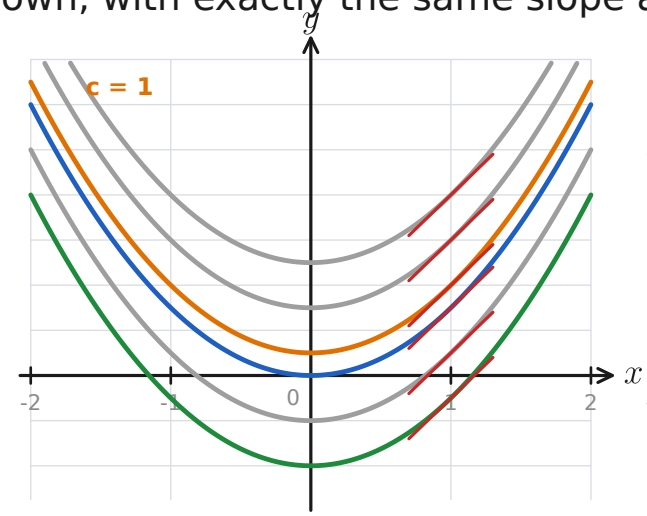
$$A'(x) = f(x)$$ ← let h shrink to 0 (Chapter 34)

**The rate the area grows is the height of the curve at the moving edge.** So $A$ is a function whose derivative is $f$: finding area **is** undoing differentiation. Check with $f(t) = t^2$: the reverse power rule says $A(x) = \frac{x^3}{3}$, and at $x = 2$, with a sliver of width 0.01:

$$\frac{A(2.01) - A(2)}{0.01} = 4.02003 \approx f(2) = 4 \checkmark$$

## + c: a whole family of answers

$3x^2$, $3x^2 + 1$ and $3x^2 - 4$ all differentiate to $6x$ (Lecture 16 calls this **degenerate**: many answers). They're the same curve lifted up or down, with exactly the same slope at every $x$:




*every curve $y = 3x^2 + c$.*
*Red: the tangents at $x = 1$ are parallel (all have slope 6).*

*So every integral gets a "+ c": the constant of integration, meaning "some number we don't know yet".*

**PROCEDURE: finding c from a point on the curve**

1. Integrate $\frac{dy}{dx}$ to get $y = F(x) + c$.
2. Substitute the given point: its $x$ into $F$, its $y$ on the left.
3. Solve for $c$.
4. Write the final equation with that number in place of $c$.

Lecture 16: $\frac{dy}{dx} = 12x^3$, and $y = 5$ when $x = 1$. The extra fact picks out one member of the family:

$$y = \int 12x^3 \, dx = 3x^4 + c \quad \longleftarrow \textit{step 1}$$

$$5 = 3(1)^4 + c \;\Rightarrow\; c = 2$$ ← steps 2–3: x = 1, y = 5

$$y = 3x^4 + 2$$ ← step 4: the specific solution

Another: a curve has gradient $3x^2 - 4$ and passes through $(2, 5)$.

$$y = x^3 - 4x + c$$ ← integrate each term

$$5 = 2^3 - 4(2) + c = 0 + c \;\Rightarrow\; c = 5$$ ← x = 2, y = 5

$$y = x^3 - 4x + 5$$

## The one exception: 1/x

Try the rule on $\int \frac{1}{x}\,\mathrm{d}x = \int x^{-1}\,\mathrm{d}x$: the power goes up to 0 and you'd divide by 0. **Impossible.** But Chapter 34 showed $\ln x$ differentiates to $\frac{1}{x}$, so:

$$\int \frac{1}{x}\,\mathrm{d}x = \ln|x| + c$$ ← |x|: the size of x, sign removed

**Slide slip:** Lecture 16 writes $\ln x + c$. The modulus matters: $\frac{1}{x}$ exists for negative $x$ too, but $\ln$ only accepts positive numbers, so we feed it $|x|$ (Chapter 17's modulus).

# Negative and fractional powers: rewrite first

**PROCEDURE: integrating anything built from powers of x**

1. Rewrite every term as $ax^n$: $\frac{3}{x^2} = 3x^{-2}$, $\sqrt{x} = x^{\frac{1}{2}}$, $\frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$, $\frac{5}{x} = 5x^{-1}$. **Expand** brackets; **split** a fraction with a single term on the bottom ($\frac{x^2+1}{x} = x + \frac{1}{x}$).
2. Integrate term by term: for $n \neq -1$, add 1 to the power and divide by the new power; a term $ax^{-1}$ gives $a\ln|x|$; a constant $k$ gives $kx$.
3. Add $+c$ and, if you like, turn the powers back into fractions and roots.
4. Check by differentiating.

$$\int \frac{3}{x^2}\,\mathrm{d}x = \int 3x^{-2}\,\mathrm{d}x = \frac{3x^{-1}}{-1} + c = -\frac{3}{x} + c$$

$-2 + 1 = -1$, $\div (-1)$

$$\int \sqrt{x}\,\mathrm{d}x = \int x^{\frac{1}{2}}\,\mathrm{d}x = \frac{x^{3/2}}{3/2} + c = \tfrac{2}{3}x^{\frac{3}{2}} + c$$

$\div \frac{3}{2}$ is $\times \frac{2}{3}$

$$\int \frac{1}{\sqrt{x}}\,\mathrm{d}x = \int x^{-\frac{1}{2}}\,\mathrm{d}x = \frac{x^{1/2}}{1/2} + c = 2\sqrt{x} + c$$

$-\frac{1}{2} + 1 = \frac{1}{2}$

$$\int (x+1)^2\,\mathrm{d}x = \int (x^2 + 2x + 1)\,\mathrm{d}x = \frac{x^3}{3} + x^2 + x + c$$

expand first

$$\int \frac{x^2+1}{x}\,\mathrm{d}x = \int \left(x + \frac{1}{x}\right)\mathrm{d}x = \frac{x^2}{2} + \ln|x| + c$$

split, then $1/x \to \ln$

## ✗ Common mistakes

✗ $\displaystyle\int \frac{3}{x^2}\,\mathrm{d}x = \frac{3}{x^3/3}$

rewrite as a power; never integrate the bottom on its own

✓ $\displaystyle\int 3x^{-2}\,\mathrm{d}x = -3x^{-1} + c$

✗ $\displaystyle\int (2x+1)^2\,\mathrm{d}x = \frac{(2x+1)^3}{3}$

expand first: the answer is $\frac{4}{3}x^3 + 2x^2 + x + c$ (Chapter 37 has a shortcut)

✓ $\displaystyle\int (4x^2 + 4x + 1)\,\mathrm{d}x$

✗ $\displaystyle\int x^{-1}\,\mathrm{d}x = \frac{x^0}{0}$

$n = -1$ is the one exception

✓ $\displaystyle\int x^{-1}\,\mathrm{d}x = \ln|x| + c$

# Standard integrals and the easy rules

**RULE: standard integrals**

Each one is a derivative from Chapter 34 read backwards. Add $+c$ to every indefinite integral.

**Sums and differences**: integrate term by term. **A number in front** waits outside: $\int kf(x)\,\mathrm{d}x = k\int f(x)\,\mathrm{d}x$.

**Products and quotients have no such rule**: expand or split them first (previous section), or use Chapter 37's methods.

Lecture 16's list (angles in radians):

| function | integral | function | integral |
|---|---|---|---|
| $k$ | $kx$ | $\sin x$ | $-\cos x$ |
| $x$ | $\frac{1}{2}x^2$ | $\cos x$ | $\sin x$ |
| $x^n$ | $\frac{1}{n+1}x^{n+1}$ | $\sin kx$ | $-\frac{1}{k}\cos kx$ |
| $\frac{1}{x}$ | $\ln\lvert x\rvert$ | $\cos kx$ | $\frac{1}{k}\sin kx$ |
| $e^x$ | $e^x$ | $\tan x$ | $\ln\lvert\sec x\rvert$ |
| $e^{kx}$ | $\frac{1}{k}e^{kx}$ | | |

About $\tan x$. Lectures 16 and 17 list $\int \tan x\,\mathrm{d}x = \ln(\sec x) + c$, where $\sec x = \frac{1}{\cos x}$. As with $\frac{1}{x}$, write $\ln\lvert\sec x\rvert$ so that it also works where $\cos x$ is negative. (Check: by the chain rule, $\ln(\sec x)$ differentiates to $\frac{\sec x\tan x}{\sec x} = \tan x$.)

The easy rules (Lecture 16). Integrate a sum or difference term by term, and pull constant multiples out front. All the separate $c$'s merge into one:

$$\int (x^2 + x^3)\,\mathrm{d}x = \frac{x^3}{3} + \frac{x^4}{4} + c \quad \longleftarrow \text{term by term}$$

$$\int 3x^4\,\mathrm{d}x = 3\int x^4\,\mathrm{d}x = \frac{3x^5}{5} + c \quad \longleftarrow \text{the 3 waits outside}$$

Lecture 16's example: $\int (4x - 2x^2 + 8)\,\mathrm{d}x$.

$$= \frac{4x^2}{2} - \frac{2x^3}{3} + 8x + c = 2x^2 - \frac{2x^3}{3} + 8x + c$$

each term: power up, ÷ new power

## Definite integrals: top minus bottom

**PROCEDURE: evaluating a definite integral**

1. Integrate (rewrite first if needed) and write the result in square brackets with the limits: $\big[F(x)\big]_a^b$. Leave out the $+c$: it would cancel.
2. Substitute the **top** limit $b$: $F(b)$.
3. Substitute the **bottom** limit $a$: $F(a)$. Use brackets; negative limits cause most mistakes.
4. Subtract: $F(b) - F(a)$. The answer is a number (a signed area), not a formula.

The square brackets mean "this, evaluated at the top minus at the bottom":

$$\int_a^b f(x)\,\mathrm{d}x = \Big[F(x)\Big]_a^b = F(b) - F(a)$$

F is any integral of f

The pool's exact answer: the area under $x^2$ from 1 to 4, which the strips were chasing:

$$\int_1^4 x^2\,\mathrm{d}x = \Big[\frac{x^3}{3}\Big]_1^4 = \frac{64}{3} - \frac{1}{3} = \frac{63}{3} = 21$$

← ✓ the strips' target

Lecture 16: $\int_1^3 2x^5\,\mathrm{d}x$.

$$= 2\Big[\frac{x^6}{6}\Big]_1^3 = \Big[\frac{x^6}{3}\Big]_1^3 = \frac{729}{3} - \frac{1}{3} = 243 - \tfrac{1}{3} = 242.\dot{6}$$

Lecture 16's your turn: $\int_{-1}^1 (3x^2 + 7)\,\mathrm{d}x$.

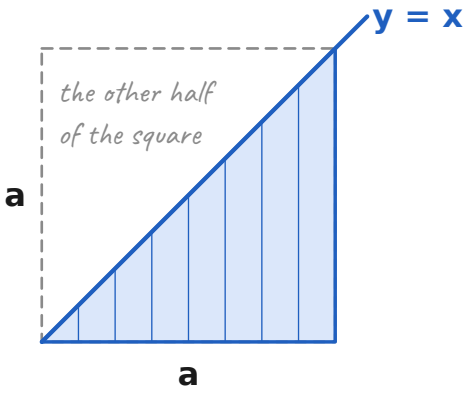
$$= \Big[x^3 + 7x\Big]_{-1}^{1} = (1+7) - (-1-7) = 8 + 8 = 16$$

*careful with the minus signs*

**Slide slip:** Lecture 16's general limits formula ends with $\frac{a}{n+1}(b^{n+1} - c^{n+1})$ for limits $b$ (bottom) to $c$ (top). It's the other way round: top minus bottom, $\frac{a}{n+1}(c^{n+1} - b^{n+1})$, as its own line above it says.

## Where the ÷2 and ÷3 come from: a triangle and a pyramid

The rule "power up by one, divide by the new power" can be checked with shapes you already know. Area under $y = x$ from 0 to $a$: the strips build a triangle, and a triangle is **half** of the $a$ by $a$ square:

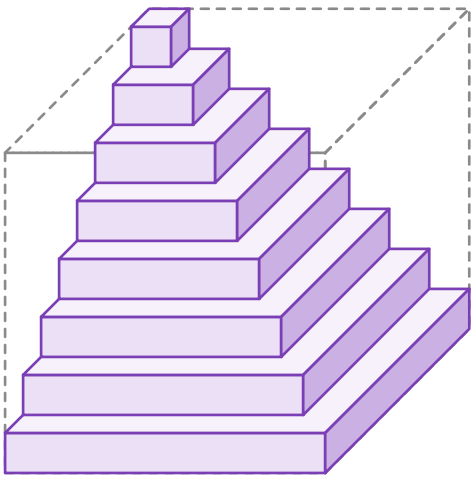



*triangle = half of a × a:*

$$\int_0^a x\,\mathrm{d}x = \left[\frac{x^2}{2}\right]_0^a = \frac{a^2}{2}$$

*the ÷2 IS the half*

Area under $y = x^2$ from 0 to $a$. Each strip is $x^2$ tall and $\mathrm{d}x$ wide. Now turn every strip into a **square slab**: $x$ by $x$, and $\mathrm{d}x$ thick (same number, $x^2\,\mathrm{d}x$). Stack the slabs, biggest at the bottom, and they build a **pyramid** with an $a$ by $a$ base and height $a$:



*dashed: the whole a × a × a cube*

*Three of these pyramids fit together to fill the whole cube (a famous fact, known to Euclid). So the pyramid is a THIRD of $a^3$:*

$$\int_0^a x^2\,\mathrm{d}x = \left[\frac{x^3}{3}\right]_0^a = \frac{a^3}{3}$$

*the ÷3 IS the third*

With 8 chunky slabs the staircase is a bit more than a third (0.398 of the cube); with 100 thin ones it's 0.338; with infinitely thin ones, exactly $\frac{1}{3}$. That's the integral doing its job. The pattern continues past what we can draw: $\int x^3 \, \mathrm{d}x$ is a 4D pyramid, a quarter of a 4D cube, $\frac{x^4}{4}$.

## Area below the axis counts as negative

**RULE: areas that cross the axis**

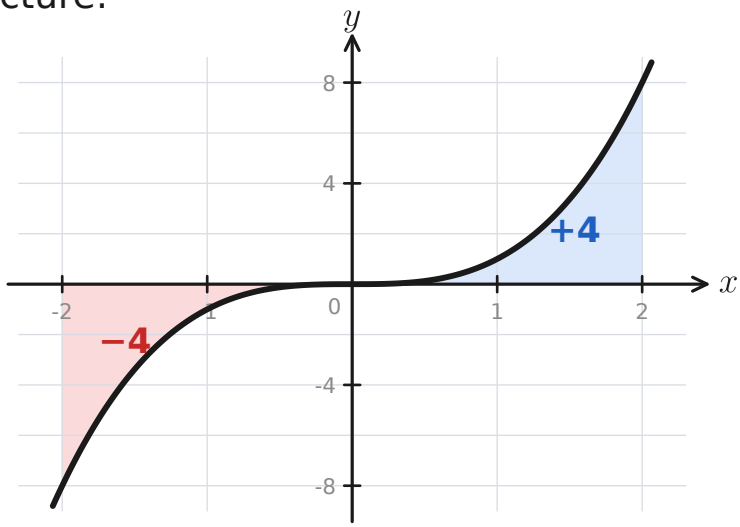
A definite integral counts area **above** the $x$-axis as positive and area **below** it as negative, so pieces on opposite sides cancel.

To find the **total area** between a curve and the axis: solve $f(x) = 0$ to find where the curve crosses the axis between the limits.

Integrate over each piece separately (limit to crossing, crossing to crossing, crossing to limit).

Add the **sizes** of the pieces (ignore their minus signs).

Lecture 16: $\int_{-2}^{2} x^3 \, \mathrm{d}x = \left[\frac{x^4}{4}\right]_{-2}^{2} = \frac{16}{4} - \frac{16}{4} = 0$. Zero area? Look at the picture:




*Area above the axis counts +, area below counts −. Here they cancel exactly.*

*If you want the total shaded area, split at the crossing and add the sizes: 4 + 4 = 8.*

**Take care interpreting them** (Lecture 16): a velocity graph dipping below the axis means going backwards, and the integral gives your net displacement (Chapter 32), not the distance run.

**Worked example: the crossing point isn't given.** Find the total area between $y = x^2 - x - 2$ and the $x$-axis from $x = 0$ to $x = 3$.

$$x^2 - x - 2 = 0 \;\Rightarrow\; (x-2)(x+1) = 0 \;\Rightarrow\; x = 2 \text{ or } x = -1$$

*solve f(x) = 0*

Only $x = 2$ lies between 0 and 3, so split there. Let $F(x) = \frac{x^3}{3} - \frac{x^2}{2} - 2x$:

$$\int_0^2 f(x)\,\mathrm{d}x = F(2) - F(0) = \left(\tfrac{8}{3} - 2 - 4\right) - 0 = -\tfrac{10}{3}$$

*below the axis*

$$\int_2^3 f(x)\,\mathrm{d}x = F(3) - F(2)$$ ← *above the axis*

$$= \left(9 - \tfrac{9}{2} - 6\right) - \left(-\tfrac{10}{3}\right) = -\tfrac{3}{2} + \tfrac{10}{3} = \tfrac{11}{6}$$

$$\text{total area} = \tfrac{10}{3} + \tfrac{11}{6} = \tfrac{31}{6} \approx 5.17$$ ← *add the sizes*

Integrating straight from 0 to 3 would give $-\frac{10}{3} + \frac{11}{6} = -\frac{3}{2}$: not an area at all.

# Find the unknown

**PROCEDURE: an integral with a missing value**

1. Do the integral with the unknown letter ($k$, $a$…) treated as a number, exactly as usual.
2. Set the result equal to the value you are given. This gives an equation in the unknown.
3. Solve it. If you get two answers, keep only the one that fits the question (e.g. $k > 1$).
4. Check by putting your answer back into the integral.

Unknown limit: $\int_1^k 2x\,\mathrm{d}x = 8$ with $k > 1$. Find $k$.

$$\left[x^2\right]_1^k = k^2 - 1$$ ← integrate, top minus bottom

$$k^2 - 1 = 8 \;\Rightarrow\; k^2 = 9 \;\Rightarrow\; k = 3$$ ← $k = -3$ is ruled out by $k > 1$

Unknown constant: $\int_0^2 (ax + 1)\,\mathrm{d}x = 10$. Find $a$.

$$\left[\tfrac{a}{2}x^2 + x\right]_0^2 = 2a + 2$$ ← $a$ is just a number

$$2a + 2 = 10 \;\Rightarrow\; a = 4$$

Unknown $c$ with a negative power: $\frac{\mathrm{d}y}{\mathrm{d}x} = 4x - \frac{3}{x^2}$, and the curve passes through $(1, 6)$.

$$y = \int \left(4x - 3x^{-2}\right)\mathrm{d}x = 2x^2 + 3x^{-1} + c = 2x^2 + \frac{3}{x} + c$$

rewrite first

$$6 = 2 + 3 + c \;\Rightarrow\; c = 1$$ ← $x = 1$, $y = 6$

$$y = 2x^2 + \frac{3}{x} + 1$$

# Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

| | |
|---|---|
| **Reverse power rule** | $\int ax^n\,dx = \dfrac{a}{n+1}x^{n+1} + c \quad (n \neq -1)$<br>power up by 1, ÷ new power |
| **Constant** | $\int k\,dx = kx + c$ |
| **The exception** | $\int \dfrac{1}{x}\,dx = \ln\lvert x\rvert + c$ |
| **Rewrite first** | $\dfrac{3}{x^2} = 3x^{-2}, \quad \sqrt{x} = x^{\frac{1}{2}}, \quad \dfrac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$<br>expand brackets; split fractions over a single term |
| **Sums and multiples** | $\int (f \pm g)\,dx = \int f\,dx \pm \int g\,dx, \qquad \int kf\,dx = k\int f\,dx$<br>no such rule for products or quotients |
| **e and trig** | $\int e^{kx}dx = \frac{1}{k}e^{kx}, \ \int \sin kx\,dx = -\frac{1}{k}\cos kx, \ \int \cos kx\,dx = \frac{1}{k}\sin kx$ |
| **tan** | $\int \tan x\,dx = \ln\lvert\sec x\rvert + c$<br>the slides write $\ln(\sec x)$ |
| **Finding c** | integrate, substitute the known point, solve for $c$ |
| **Definite integral** | $\int_a^b f(x)\,dx = F(b) - F(a)$<br>top minus bottom; no $+c$ |
| **Total area** | solve $f(x) = 0$, integrate each piece, add the sizes |
| **Find the unknown** | integrate with the letter as a number, set equal to the given value, solve |
| **Trapezium rule** | $\dfrac{h}{2}\big(y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n\big), \quad h = \dfrac{b-a}{n}$ |

# Your turn

Check every indefinite answer by differentiating it. Answers in grey.

## Trapezium rule

1. Estimate the area under $y = x^2$ from 0 to 2 with 2 strips, then 4 strips. (True: 8/3.)

1. 3 and 2.75

## Indefinite integrals

2. $\int 6x^2\,dx$
3. $\int (3x^2 - 4x + 5)\,dx$
4. $\int (2/x)\,dx$
5. $\int e^{(3x)}\,dx$
6. $\int 4\cos 2x\,dx$

2. $2x^3 + c$  3. $x^3 - 2x^2 + 5x + c$  4. $2\ln|x| + c$  5. $\frac{1}{3}e^{(3x)} + c$  6. $2\sin 2x + c$

## Find the curve

7. $dy/dx = 6x + 2$ and the curve passes through (1, 10).

7. $y = 3x^2 + 2x + 5$

## Definite integrals

8. $\int$ from 0 to 3 of $x^2\,dx$
9. $\int$ from 1 to 2 of $(4x^3 - 1)\,dx$
10. The shaded area between $y = x^3$ and the x axis from $-1$ to 1

8. 9  9. 14  10. ½ (each side is ¼; the integral itself is 0)

## Rewrite first

11. $\int 4/x^3\,dx$
12. $\int 6\sqrt{x}\,dx$
13. $\int (x + 2)^2\,dx$
14. $\int (x^3 - 5)/x\,dx$
15. Total area between $y = x^2 - 1$ and the x axis from $x = 0$ to $x = 2$

11. $-2/x^2 + c$ ($4x^{-3} \to 4x^{-2}/(-2)$)  12. $4x^{(3/2)} + c$  13. $x^3/3 + 2x^2 + 4x + c$  14. $x^3/3 - 5\ln|x| + c$  15. 2 (crosses at $x = 1$: pieces $-\frac{2}{3}$ and 4/3)

## Find the unknown

16. $dy/dx = 4x + 3$ and the curve passes through $(1, 2)$. Find $y$.

17. $\int$ from 1 to $k$ of $2x\,dx = 15$, with $k > 1$. Find $k$.

18. $\int$ from 0 to 3 of $(ax + 2)\,dx = 15$. Find $a$.

19. $\int$ from 0 to $k$ of $3x^2\,dx = 64$. Find $k$.

20. $f'(x) = 6x^2 - 2/x^2$ and $f(1) = 5$. Find $f(x)$.

16. $y = 2x^2 + 3x - 3$ ($2 + 3 + c = 2$)  17. $k = 4$ ($k^2 - 1 = 15$)  18. $a = 2$ ($9a/2 + 6 = 15$)  19. $k = 4$ ($k^3 = 64$)  20. $f(x) = 2x^3 + 2/x + 1$ ($2 + 2 + c = 5$)